Stats Final Enzyme Data

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Target Audience:

Undergrads and fellow grad students

Rationale, Background info, Questions:

Moray eels have the ability to eat really large and infrequently (though commonly do not) and can fast for extended periods of time, making them an interesting model of fasting ectothermic fish. There is evidence of regionality in enzymatic activity in many other species, meaning that within different areas of the stomach and intestine, different enzymes can have different levels of activity, though not much has been done on fishes, especially nonfishery species. My target enzymes for this analysis are all the enzymes in my data set.

Question 1: Is there gut regionality of enzymatic activity for the California Moray Eel (Gymnothorax mordax)?

Chitin is also a very important polysaccharride used in crustacean shells as protection from other organisms. One of its uses is that it can be difficult to digest. While there are some species of morays that are capable of crunching their prey, California moray eels swallow their prey whole, and therefore have to get past a chitinous shell to get to the goodies inside the crustacean bodies. Since they are also capable of fasting for extended periods of time its likely they upregulate their enzymatic activity in response to specific prey types and downregulate to save energy when they are not actively digesting prey with specific complex molecules. This occurs very quickly in between meals in organisms like pythons who are known to eat big and fast, but not known to occur in between meals organisms that eat frequently, with some organisms not exhibiting any upregulation in response to specific prey regardless of what they eat. I want to see if the eels are capable of upregulating specific enzymes in response to specific prey during the meal. The specific enzymes I examined that are directly associated with the breakdown of chitin are chitinase and NAGase, which are the targets for the following chitin question.

Question 2: Is chitin degrading enzymatic activity higher in California Moray Eels who fed on crustaceans recently?

Approach:

I completed enzymatic assays on 6 different regions of the GI tract of 10 wild caught eels without controlling for diet, and recorded exactly what was found in their GI tract (all of them had prey present). My statistical approach to this is first creating visualizations of the data in the form of bar charts. This does not control for random effects or factors such as individual or chitin for regionality and regionality for chitin, so I tried to account for this by creating models that accounted for all of the factors I had brought in.

I completed some data wrangling on the data sets, then created simple box plots to help visualize the data. Then, I created a model for each independent enzyme with the following factors:

Enzymatic Activity in the specific gut region

Whether there was a crustacean prey in the gut

Random Factor: Who the individual was (this would help account for any individuals who might just be having higher activity in general)

By creating models using brms of both the activity for regions of the gut and the crustacean presence in the moray guts, I will be able to answer both my question of regionality and my question of whether there's a difference in specific enzyme activity levels based on prey. By creating a for loop that created a model for each enzyme, I was able to examine the results for both of my questions for each enzyme, not just the targets of the question, though I still focused on NAGase and chitinase for my chitin response question. This let me see how enzymes I wouldn't have thought to look at reacted to the presence of chitin.

Highlights of the project

Q1: I found some regionality of certain enzymes based on gut region.

Chitinase was more active in the anterior of the stomach, pepsin was more active in the middle and posterior of the stomach, lipase was more active in the anterior of the intestine, and maltase and amylase were more active in the anterior and middle of the intestine. All other enzymes showed very little or no regionality at all. It is very interesting that only some of them showed clear regionality and some others didn't, which begs further research.

Q2: While I didn't find that the activity was significantly higher when there were crustaceans present in the gut, both chitinase and NAGase had a much higher range of activities when crustaceans were present. To give this a possible reason in ecological context, all eels eat crustaceans, but some eat more crustaceans more often than others. This would result in individuals who eat them more often having higher chitin degrading enzyme activity than those who only eat crustaceans occasionally. This would mean that the species can upregulate specific enzyme activity over time when they consistently eat crustacean meals in order to

break down specific molecules in the food, whereas if they do not eat the meal consistently enough, they do not upregulate those enzymes.

Pepsin also resulted in a strangely large range of predicted responses for the brm model that I created for it examining the crustacean presence. While the chains of this model didn't all converge and the rhats were all higher than they should be, it is an interesting result that happened even after running the model creation twice.

All of these model results for both Q1 and Q2 should be taken with a grain of salt due to none of the models having good rhats, unconverging chains, and multiple peaks. This is likely due to the lack of overall data inputted into training the model. However, the boxplots of the data do align with the models to some extent.

Main message

We found some evidence of GI regionality of enzymatic activity for chitinase, pepsin, lipase, amylase, and maltase. We also found that morays with crustaceans present in their guts had larger ranges of chitin degrading enzymes.

Lessons learned along the way

Sometimes you gotta just work with the data you have, where if the data isn't enough to run a perfect model and get great results, you still try and keep that in mind as you interpret the results.

It can be nice to expect all code to fail the first time you run it, that way you aren't upset with it immediately failing with it inevitably does.

My computer really doesn't like running multiple models back to back

For github, make sure to have a metadata

Reading the csv files of the data with better call back names

```
stomach <- read.csv("Copy of Enzyme Data For Stats Class - Stomach Regionality.csv")
intestine<- read.csv("Copy of Enzyme Data For Stats Class - Small Intestine Regionality.csv")
upreg<-read.csv("Copy of Enzyme Data For Stats Class - Upregulation .csv")
```

Preparing Packages

```r
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.1
```

```r
library(here)
```

```
Warning: package 'here' was built under R version 4.5.2

here() starts at C:/Users/gabec/Downloads/Stats Malin Winter 2026/Final Project/BIOE-
276-Stats_Final
```

```r
library(tidyverse)
```

```
Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.1

Warning: package 'stringr' was built under R version 4.5.1

Warning: package 'forcats' was built under R version 4.5.1

Warning: package 'lubridate' was built under R version 4.5.2

-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.2
v lubridate 1.9.4     v tibble    3.3.1
v purrr     1.1.0     v tidyr     1.3.1

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(brms)
```

```
Warning: package 'brms' was built under R version 4.5.2

Loading required package: Rcpp

Warning: package 'Rcpp' was built under R version 4.5.1

Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').

Attaching package: 'brms'

The following object is masked from 'package:stats':

    ar
```

```r
library(ggeffects)
```

```
Warning: package 'ggeffects' was built under R version 4.5.2
```

```r
library(modelbased)
```

```
Warning: package 'modelbased' was built under R version 4.5.2


Attaching package: 'modelbased'

The following objects are masked from 'package:ggeffects':

    collapse_by_group, pool_predictions, residualize_over_grid
```

```r
library(faraway)
```

```
Warning: package 'faraway' was built under R version 4.5.2
```

```
Attaching package: 'faraway'

The following object is masked from 'package:brms':

    epilepsy
```

```r
head(intestine)
```

```
  Sample_ID ID Crustacean.Present   GI_tract_location        Amylase    Maltase
1       EA1  1                Yes  Anterior Intestine Not Detectable 0.56005594
2       EM1  1                Yes    Middle Intestine       0.324786 0.41172840
3       EP1  1                Yes Posterior Intestine       0.757977 0.45082305
4       EA2  2                 No  Anterior Intestine       0.231909 0.27952675
5       EM2  2                 No    Middle Intestine       0.487749 0.07983539
6       EP2  2                 No Posterior Intestine Not Detectable 0.04453704
            Lipase       NAG        AP      Trypsin Amino_Peptidase
1    0.05594225901  64.49046 156.25095  0.767357747       0.6789927
2    0.01131218769 105.17670 223.13148  1.123875433       0.2491349
3   0.008645034001 343.74618 164.14013   2.52560073       0.9326221
4    0.01837231267  54.68058 100.88953  0.434544406       0.4347559
5   0.008136497856  70.37718  85.39748 0.8381391772       1.0666667
6 0.00002844161178  66.55975 135.81755 0.5627527874       0.4159477
```

```r
head(stomach)
```

```
  Sample_ID ID Crustacean.Present GI_tract_location      Chitinase
1       EF1  1                Yes    Stomach Fundus Not Detectable
2       EB1  1                Yes      Stomach Body Not Detectable
3      EAN1  1                Yes    Stomach Antrum Not Detectable
4       EF2  2                 No    Stomach Fundus        0.00336
5       EB2  2                 No      Stomach Body  0.05974320988
6      EAN2  2                 No    Stomach Antrum Not Detectable
          Pepsin
1   0.8008296296
2    2.261111111
3 Not Detectable
4   0.1705777778
5     3.85037037
6 Not Detectable
```

```r
head(upreg)
```

```
  Sample_ID Crustacean.Present      Chitinase    NAGase
1       EF1                Yes Not Detectable  64.49046
2       EB1                Yes Not Detectable 105.17670
3      EAN1                Yes Not Detectable 343.74618
4       EF2                 No        0.00336  54.68058
5       EB2                 No  0.05974320988  70.37718
6      EAN2                 No Not Detectable  66.55975
```

Reformatting tibbles to better work with our analysis

Turning Not Detectables into NAs so the code doesn't read them as characters.

(Note "Not Detectables" are NOT 0s in the data, it means that the assay itself could not detect it and should be listed as such)

```r
intestine <- intestine %>%
  mutate(across(c(Amylase:Amino_Peptidase),
                ~ as.numeric(.)))
```

```
Warning: There were 3 warnings in `mutate()`.
The first warning was:
i In argument: `across(c(Amylase:Amino_Peptidase), ~as.numeric(.))`.
Caused by warning:
! NAs introduced by coercion
i Run `dplyr::last_dplyr_warnings()` to see the 2 remaining warnings.
```

```r
stomach <- stomach %>%
  mutate(across(c(Chitinase:Pepsin),
                ~ as.numeric(.)))
```

```
Warning: There were 2 warnings in `mutate()`.
The first warning was:
i In argument: `across(c(Chitinase:Pepsin), ~as.numeric(.))`.
Caused by warning:
! NAs introduced by coercion
i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```r
upreg<-upreg %>%
  mutate(across(c(Chitinase:NAGase),
                ~ as.numeric(.)))
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `across(c(Chitinase:NAGase), ~as.numeric(.))`.
Caused by warning:
! NAs introduced by coercion
```

Changing the tibbles to long format

```r
intestine_long <- intestine %>%
  pivot_longer(
    cols = c(Amylase, Maltase, Lipase, NAG, AP, Trypsin, Amino_Peptidase),
    names_to = "Enzyme",
    values_to = "Activity")
stomach_long <- stomach %>%
  pivot_longer(
    cols = c(Chitinase,Pepsin),
    names_to = "Enzyme",
    values_to = "Activity")
```

```r
upreg_long <- upreg %>%
  pivot_longer(
    cols = c(Chitinase,NAGase),
    names_to = "Enzyme",
    values_to = "Activity")
```

Changing the names of the location to remove naming redundancies in the figures.

```r
stomach_long <- stomach_long %>%
  mutate(GI_tract_location = str_sub(GI_tract_location, 9))
```

```r
intestine_long_n <- intestine_long %>%
  mutate(GI_tract_location = word(GI_tract_location, 1))
```

Making the box plots

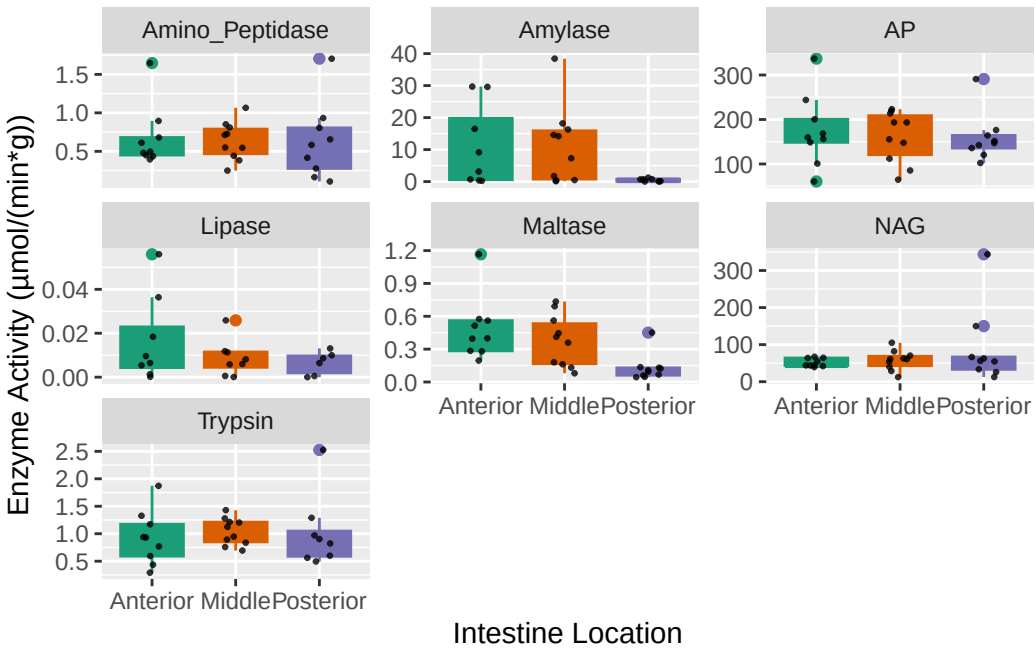
Note: NAs will give warnings here and won't be displayed

Stomach Regionality

```
ggplot(intestine_long_n, aes(x = factor(GI_tract_location, levels = c("Anterior", "Middle", 
  geom_boxplot() +
  geom_point(position = position_jitter(width = 0.15),
             size = .5,
             alpha = 0.8,
             color = "black")+
  facet_wrap(~ Enzyme, scales = "free_y")+
  labs(x="Intestine Location", y="Enzyme Activity (µmol/(min*g))")+
  theme(legend.position = "none")+
  scale_fill_brewer(palette = "Dark2") +
  scale_color_brewer(palette = "Dark2")
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 15 rows containing missing values or values outside the scale range
(`geom_point()`).
```

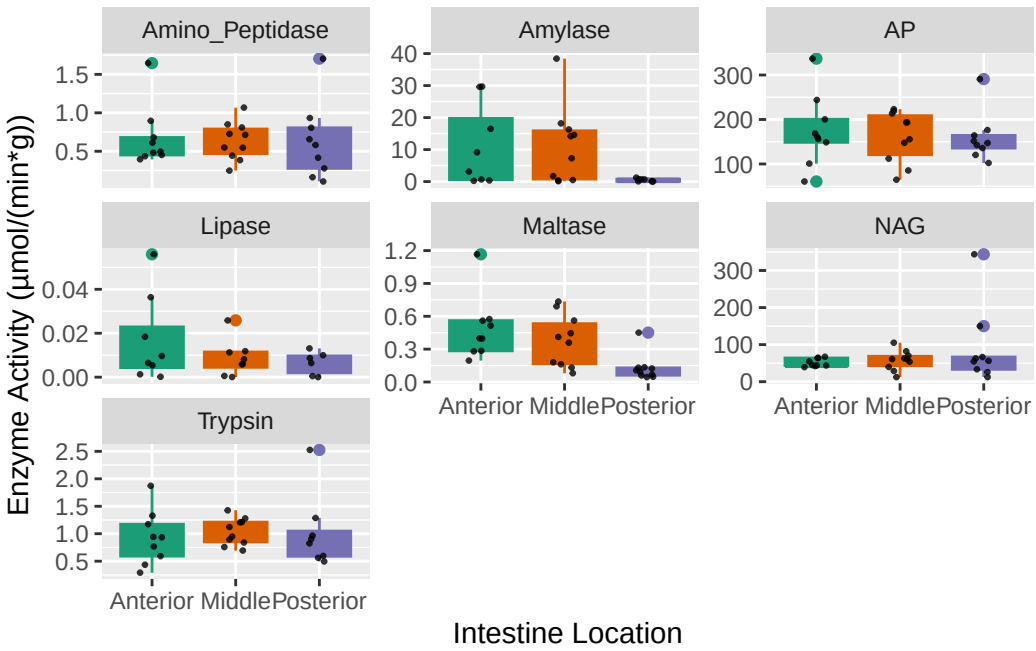


Intestine Regionality

```
ggplot(intestine_long_n, aes(x = factor(GI_tract_location, levels = c("Anterior", "Middle", "
  geom_boxplot() +
  geom_point(position = position_jitter(width = 0.15),
             size = .5,
             alpha = 0.8,
             color = "black")+
  facet_wrap(~ Enzyme, scales = "free_y")+
  labs(x="Intestine Location", y="Enzyme Activity (µmol/(min*g))")+
  theme(legend.position = "none")+
  scale_fill_brewer(palette = "Dark2") +
  scale_color_brewer(palette = "Dark2")
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

```
Warning: Removed 15 rows containing missing values or values outside the scale range
(`geom_point()`).
```

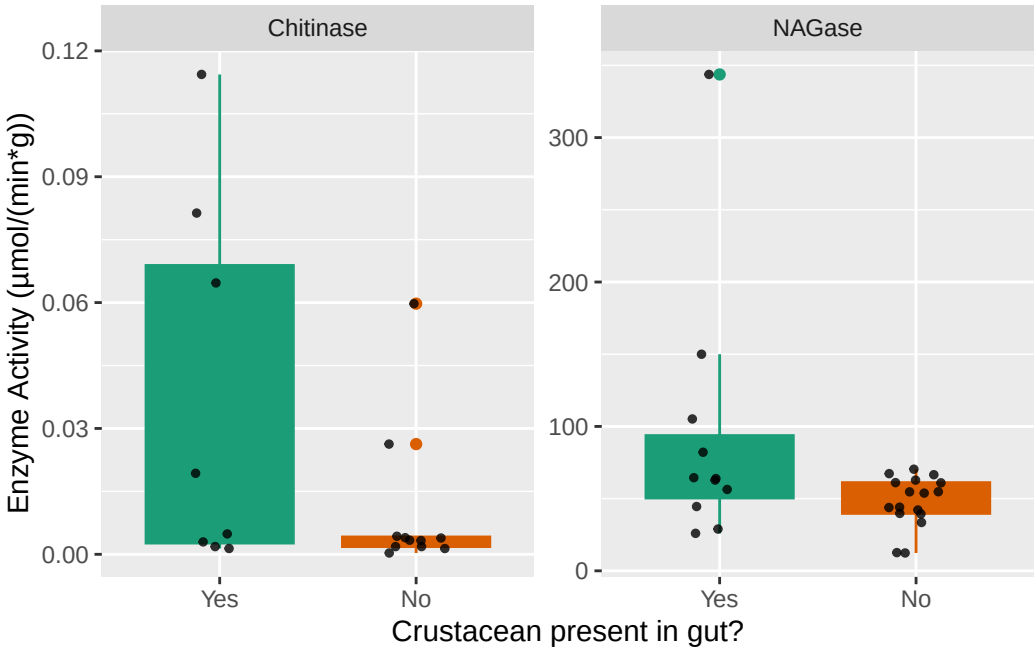


Upregulation of chitin degrading enzymes based on chitinous prey presense?

```
ggplot(upreg_long, aes(x = factor(Crustacean.Present, levels = c("Yes", "No")), y = Activity
  geom_boxplot() +
  facet_wrap(~ Enzyme, scales = "free_y")+
  geom_point(position = position_jitter(width = 0.15),
             size = 1,
             alpha = 0.8,
             color = "black")+
  labs(x="Crustacean present in gut?", y="Enzyme Activity (µmol/(min*g))")+
  theme(legend.position = "none")+
  scale_fill_brewer(palette = "Dark2", direction = -1) +
  scale_color_brewer(palette = "Dark2", direction = -1)
```

```
Warning: Removed 13 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

```
Warning: Removed 13 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Building the models for Intestinal Data and showing results

```
library(dplyr)
library(purrr)
library(brms)
dir.create("output")
```

```
Warning in dir.create("output"): 'output' already exists
```

```
# Split data by Enzyme
grouped <- intestine_long_n %>%
  group_by(Enzyme)

enzyme_data  <- group_split(grouped)
enzyme_names <- group_keys(grouped)$Enzyme

# Fit model for each enzyme
models.i <- map2(
  enzyme_data,
  enzyme_names,
```

```r
  ~ {    # Calculate enzyme-specific prior center
    priors_i <- c(
      prior(normal(log(mean(.x$Activity)), 1), class = "b"),
      prior(exponential(1), class = "sd"),
      prior(gamma(2, 0.1), class = "shape")
    )
    brm(
      data = .x,
      family = Gamma,
      Activity ~ 0 + GI_tract_location + Crustacean.Present +
        (1|ID),
      control = list(adapt_delta = .9),
       iter = 2000, warmup = 1000, chains = 4, cores = 8,
      file = paste0("output/m.intestine_", .y)
    )}
)
```

```
Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Trying to compile a simple C file

Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.2.0'
gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/gabec/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/gabec/AppData/Local/
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/gabec/AppDa
```

```
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -
I"C:/Users/gabec/AppData/Local/R/win-library/4.4/rtools45/x86_64-w64-mingw32.static.posix/inc
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/StanHeaders/include/stan
                 from <command-line>:
C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Start sampling

Warning: There were 829 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: There were 1 chains where the estimated Bayesian Fraction of Missing Information was
https://mc-stan.org/misc/warnings.html#bfmi-low

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: The largest R-hat is 1.44, indicating chains have not mixed.
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#r-hat

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tai
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess
```

```
Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
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Trying to compile a simple C file

Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.2.0'
gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/gabec/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/gabec/AppData/Local/
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/gabec/AppDat
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -
I"C:/Users/gabec/AppData/Local/R/win-library/4.4/rtools45/x86_64-w64-mingw32.static.posix/inc
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/StanHeaders/include/stan
                 from <command-line>:
C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h
  679 | #include <cmath>
      |          ^~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1

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Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.
```

```
Start sampling

Warning: There were 100 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...

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using C compiler: 'gcc.exe (GCC) 14.2.0'
gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/gabec/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/gabec/AppData/Local/
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/gabec/AppDat
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -
I"C:/Users/gabec/AppData/Local/R/win-library/4.4/rtools45/x86_64-w64-mingw32.static.posix/inc
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/StanHeaders/include/stan
                 from <command-line>:
C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h
```

```
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Start sampling

Warning in .local(object, ...): some chains had errors; consider specifying
chains = 1 to debug

here are whatever error messages were returned

[[1]]
Stan model 'anon_model' does not contain samples.

Warning: There were 1408 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and median
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and ta
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess

Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...
```

```
WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Trying to compile a simple C file

Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.2.0'
gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/gabec/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/gabec/AppData/Local/
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/gabec/AppDat
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -
I"C:/Users/gabec/AppData/Local/R/win-library/4.4/rtools45/x86_64-w64-mingw32.static.posix/inc
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/StanHeaders/include/stan
                 from <command-line>:
C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Start sampling

Warning in .local(object, ...): some chains had errors; consider specifying
chains = 1 to debug
```

```
here are whatever error messages were returned

[[1]]
Stan model 'anon_model' does not contain samples.

Warning: There were 1247 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: Examine the pairs() plot to diagnose sampling problems

Compiling Stan program...

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Trying to compile a simple C file

Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.2.0'
gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/gabec/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/gabec/AppData/Local/
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/gabec/AppDat
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -
I"C:/Users/gabec/AppData/Local/R/win-library/4.4/rtools45/x86_64-w64-mingw32.static.posix/inc
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/StanHeaders/include/stan
                 from <command-line>:
```

```
C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Start sampling

Warning in .local(object, ...): some chains had errors; consider specifying
chains = 1 to debug

here are whatever error messages were returned

[[1]]
Stan model 'anon_model' does not contain samples.

[[2]]
Stan model 'anon_model' does not contain samples.

[[3]]
Stan model 'anon_model' does not contain samples.

Warning: There were 34 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.
Warning: Examine the pairs() plot to diagnose sampling problems

Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.
```

```
Trying to compile a simple C file

Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.2.0'
gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/gabec/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/gabec/AppData/Local/
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/gabec/AppDat
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -
I"C:/Users/gabec/AppData/Local/R/win-library/4.4/rtools45/x86_64-w64-mingw32.static.posix/inc
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/StanHeaders/include/stan
                 from <command-line>:
C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Start sampling

Warning in .local(object, ...): some chains had errors; consider specifying
chains = 1 to debug

here are whatever error messages were returned

[[1]]
Stan model 'anon_model' does not contain samples.
```

```
Warning: There were 1133 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: There were 47 transitions after warmup that exceeded the maximum treedepth. Increase
https://mc-stan.org/misc/warnings.html#maximum-treedepth-exceeded

Warning: There were 1 chains where the estimated Bayesian Fraction of Missing Information was
https://mc-stan.org/misc/warnings.html#bfmi-low

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: The largest R-hat is 1.73, indicating chains have not mixed.
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#r-hat

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tai
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess

Warning: Rows containing NAs were excluded from the model.

Compiling Stan program...

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Trying to compile a simple C file
```

```
Running "C:/PROGRA~1/R/R-45~1.0/bin/x64/Rcmd.exe" SHLIB foo.c
using C compiler: 'gcc.exe (GCC) 14.2.0'
gcc  -I"C:/PROGRA~1/R/R-45~1.0/include" -DNDEBUG   -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/Rcpp/include/"  -I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/inclu
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/unsupported"  -
I"C:/Users/gabec/AppData/Local/R/win-library/4.5/BH/include" -I"C:/Users/gabec/AppData/Local/
library/4.5/StanHeaders/include/src/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/StanHeaders/include/"  -I"C:/Users/gabec/AppData/Local/R/win-
library/4.5/RcppParallel/include/" -DRCPP_PARALLEL_USE_TBB=1 -I"C:/Users/gabec/AppData/Local/
library/4.5/rstan/include" -DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -
DBOOST_PENDING_INTEGER_LOG2_HPP  -DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -
DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -D_HAS_AUTO_PTR_ETC=0  -include "C:/Users/gabec/AppDat
library/4.5/StanHeaders/include/stan/math/prim/fun/Eigen.hpp"  -std=c++1y    -
I"C:/Users/gabec/AppData/Local/R/win-library/4.4/rtools45/x86_64-w64-mingw32.static.posix/inc
O2 -Wall -std=gnu2x  -mfpmath=sse -msse2 -mstackrealign   -c foo.c -o foo.o
cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
In file included from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/
                 from C:/Users/gabec/AppData/Local/R/win-library/4.5/StanHeaders/include/stan
                 from <command-line>:
C:/Users/gabec/AppData/Local/R/win-library/4.5/RcppEigen/include/Eigen/src/Core/util/Macros.h
  679 | #include <cmath>
      |          ^~~~~~~
compilation terminated.
make: *** [C:/PROGRA~1/R/R-45~1.0/etc/x64/Makeconf:289: foo.o] Error 1

WARNING: Rtools is required to build R packages, but is not currently installed.

Please download and install the appropriate version of Rtools for 4.5.0 from
https://cran.r-project.org/bin/windows/Rtools/.

Start sampling

Warning in .local(object, ...): some chains had errors; consider specifying
chains = 1 to debug

here are whatever error messages were returned

[[1]]
Stan model 'anon_model' does not contain samples.
```

```
Warning: There were 12 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: Examine the pairs() plot to diagnose sampling problems
```

```r
names(models.i) <- enzyme_names
```

```r
print(models.i, digits = 3)
```

```
$AP

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
careful when analysing the results! We recommend running more iterations and/or
setting stronger priors.

Warning: There were 829 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 28)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~ID (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.002     0.006    0.000    0.017 1.231       13       13

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAnterior     0.017     0.037    0.005    0.185 1.186       14
GI_tract_locationMiddle       0.018     0.039    0.006    0.174 1.188       14
GI_tract_locationPosterior    0.017     0.033    0.006    0.132 1.187       14
Crustacean.PresentYes        -0.010     0.029   -0.114    0.000 1.244       12
                          Tail_ESS
GI_tract_locationAnterior        11
```

```
GI_tract_locationMiddle          11
GI_tract_locationPosterior       11
Crustacean.PresentYes            11

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    6.751     2.968    0.056   12.246 1.224       13       11

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

$Amino_Peptidase


Warning: There were 100 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup


 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 28)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~ID (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.429     0.275    0.034    1.069 1.002     1103     1400

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAnterior     1.812     0.379    1.103    2.603 1.003     1895
GI_tract_locationMiddle       1.913     0.380    1.202    2.697 1.001     1541
GI_tract_locationPosterior    1.956     0.404    1.187    2.776 1.003     1575
Crustacean.PresentYes        -0.398     0.472   -1.299    0.553 1.003     1441
                          Tail_ESS
GI_tract_locationAnterior      2264
GI_tract_locationMiddle        1849
GI_tract_locationPosterior     1989
Crustacean.PresentYes          1606
```

```
Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    3.610     1.078    1.826    6.107 1.002     2141     2138

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

$Amylase


Warning: There were 1408 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup


 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 26)
  Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 3000

Multilevel Hyperparameters:
~ID (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.064     0.064    0.002    0.240 1.012      226      280

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAnterior     0.144     0.064    0.045    0.301 1.009      369
GI_tract_locationMiddle       0.142     0.063    0.047    0.297 1.008      326
GI_tract_locationPosterior    2.545     1.072    0.911    5.121 1.001      789
Crustacean.PresentYes         0.005     0.086   -0.149    0.186 1.007      469
                          Tail_ESS
GI_tract_locationAnterior       240
GI_tract_locationMiddle         209
GI_tract_locationPosterior      586
Crustacean.PresentYes           408

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    0.611     0.158    0.350    0.969 1.005      645      670
```

```
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

$Lipase

Warning: There were 1247 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 22)
  Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 3000

Multilevel Hyperparameters:
~ID (Number of levels: 9)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)   36.999    31.506    1.414  116.576 1.009      488      654

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAnterior   115.170    46.274   39.374  216.692 1.010      452
GI_tract_locationMiddle     171.417    65.601   64.698  322.620 1.010      652
GI_tract_locationPosterior  230.145    97.951   76.556  440.184 1.002      960
Crustacean.PresentYes       -37.758    58.947 -159.096   77.786 1.005      637
                          Tail_ESS
GI_tract_locationAnterior       635
GI_tract_locationMiddle         503
GI_tract_locationPosterior      936
Crustacean.PresentYes           880

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    0.607     0.172    0.319    0.998 1.003      925      875

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
$Maltase

Warning: There were 34 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 29)
  Draws: 1 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 1000

Multilevel Hyperparameters:
~ID (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.639     0.510    0.038    1.839 1.002      205      268

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAnterior     3.065     0.722    1.695    4.466 1.003      476
GI_tract_locationMiddle       3.672     0.765    2.250    5.231 1.003      490
GI_tract_locationPosterior    8.827     1.631    5.799   12.409 1.004      864
Crustacean.PresentYes        -1.378     0.925   -3.045    0.721 1.007      332
                          Tail_ESS
GI_tract_locationAnterior      381
GI_tract_locationMiddle        416
GI_tract_locationPosterior     543
Crustacean.PresentYes          161

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    2.933     0.827    1.647    4.865 0.999      559      373

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

$NAG

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
careful when analysing the results! We recommend running more iterations and/or
```

```
setting stronger priors.

Warning: There were 1133 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 28)
  Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 3000

Multilevel Hyperparameters:
~ID (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.081     0.115    0.002    0.340 1.728        5       11

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAnterior     0.058     0.051    0.017    0.170 1.587        5
GI_tract_locationMiddle       0.055     0.049    0.016    0.163 1.589        5
GI_tract_locationPosterior    0.052     0.051    0.012    0.165 1.579        5
Crustacean.PresentYes        -0.082     0.112   -0.301    0.003 1.587        5
                          Tail_ESS
GI_tract_locationAnterior        19
GI_tract_locationMiddle          42
GI_tract_locationPosterior       15
Crustacean.PresentYes            30

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    3.232     2.474    0.105    7.816 1.706        5       13

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

$Trypsin

Warning: There were 12 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
```

```
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 27)
  Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 3000

Multilevel Hyperparameters:
~ID (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.153     0.128    0.005    0.463 1.002     1159     1313

Regression Coefficients:
                          Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAnterior     1.225     0.199    0.858    1.621 1.001     1940
GI_tract_locationMiddle       1.114     0.188    0.743    1.501 1.001     1657
GI_tract_locationPosterior    1.123     0.202    0.756    1.521 1.001     1924
Crustacean.PresentYes        -0.233     0.211   -0.625    0.183 1.001     1435
                          Tail_ESS
GI_tract_locationAnterior      1746
GI_tract_locationMiddle        1617
GI_tract_locationPosterior     1847
Crustacean.PresentYes          1636

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    5.176     1.477    2.673    8.364 1.000     2710     2067

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
Amylase_pred <- predict_response(models.i[[1]]) #, bias_correction = TRUE)
```

```
You are calculating adjusted predictions on the population-level (i.e.
  `type = "fixed"`) for a *generalized* linear mixed model.
  This may produce biased estimates due to Jensen's inequality. Consider
  setting `bias_correction = TRUE` to correct for this bias.
  See also the documentation of the `bias_correction` argument.
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
glmmTMB was built with TMB package version 1.9.17
Current TMB package version is 1.9.18
Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling'

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
Maltase <- predict_response(models.i[[2]]) #, bias_correction = TRUE)
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
Lipase_pred <- predict_response(models.i[[3]]) #, bias_correction = TRUE)
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
NAGase_pred <- predict_response(models.i[[4]]) #, bias_correction = TRUE)
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
APase_pred <- predict_response(models.i[[5]]) #, bias_correction = TRUE)
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
Trypsin_pred <- predict_response(models.i[[6]]) #, bias_correction = TRUE)
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
A.peptidase_pred <- predict_response(models.i[[7]]) #, bias_correction = TRUE)
```

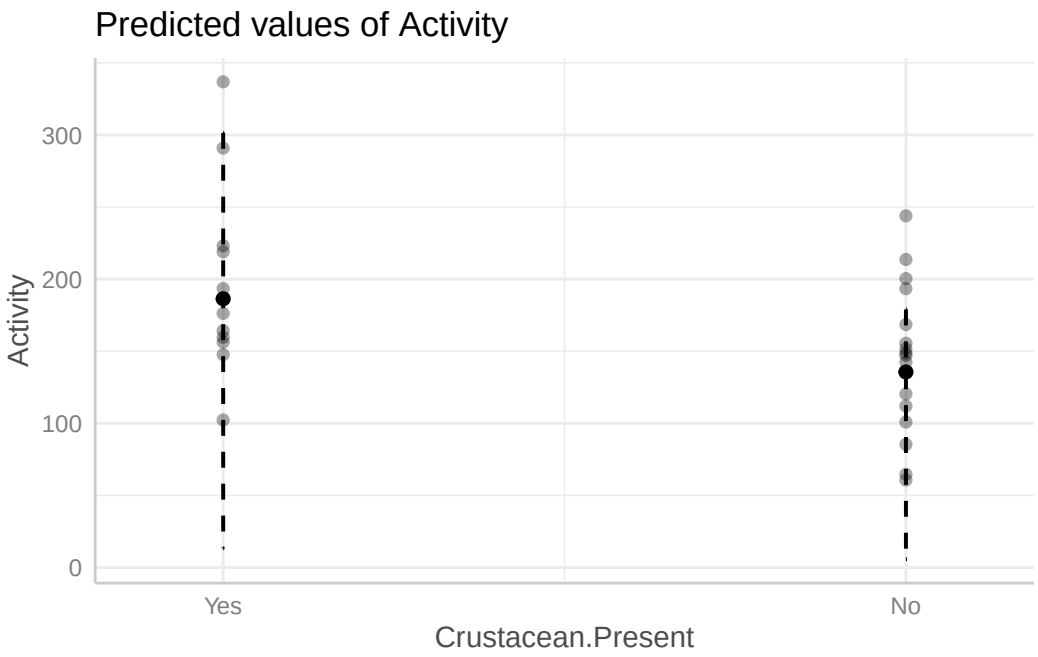
```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

Plotting the model predictions as ranges in a format similar to the earlier box plot visualizations

```r
plot(Amylase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.

$GI_tract_location
```

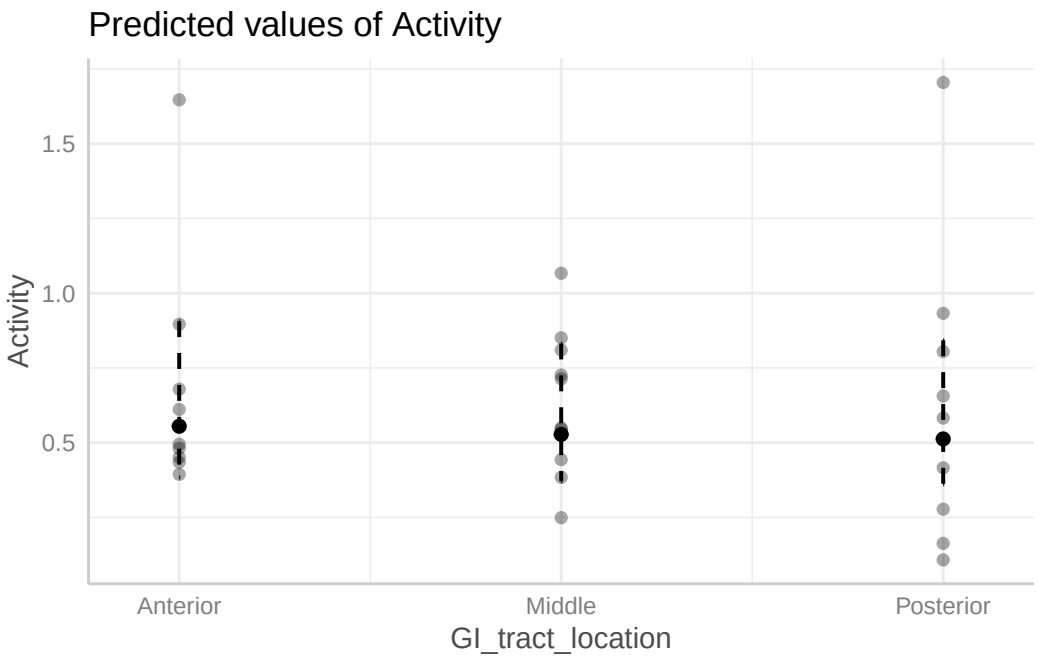
```
$Crustacean.Present
```



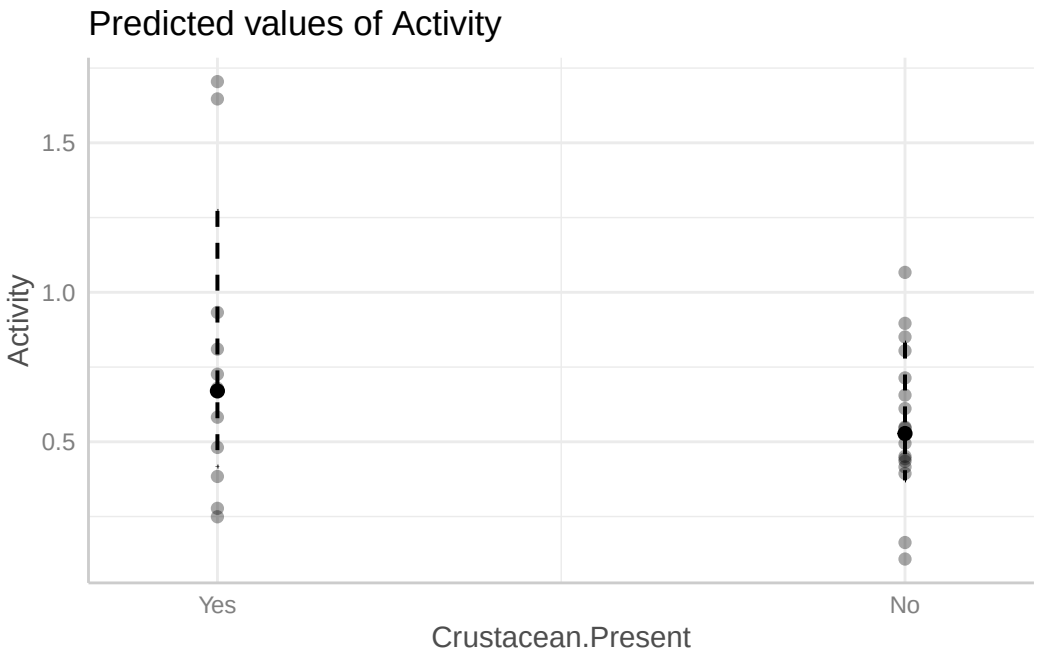
```r
plot(Maltase, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```

```
$GI_tract_location
```

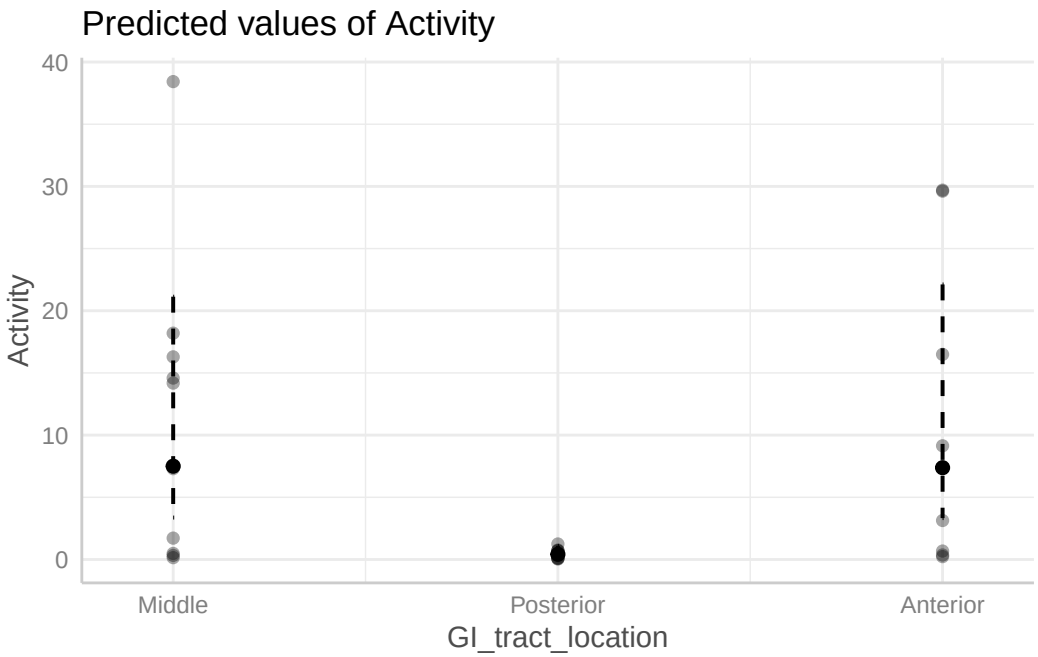



```
$Crustacean.Present
```

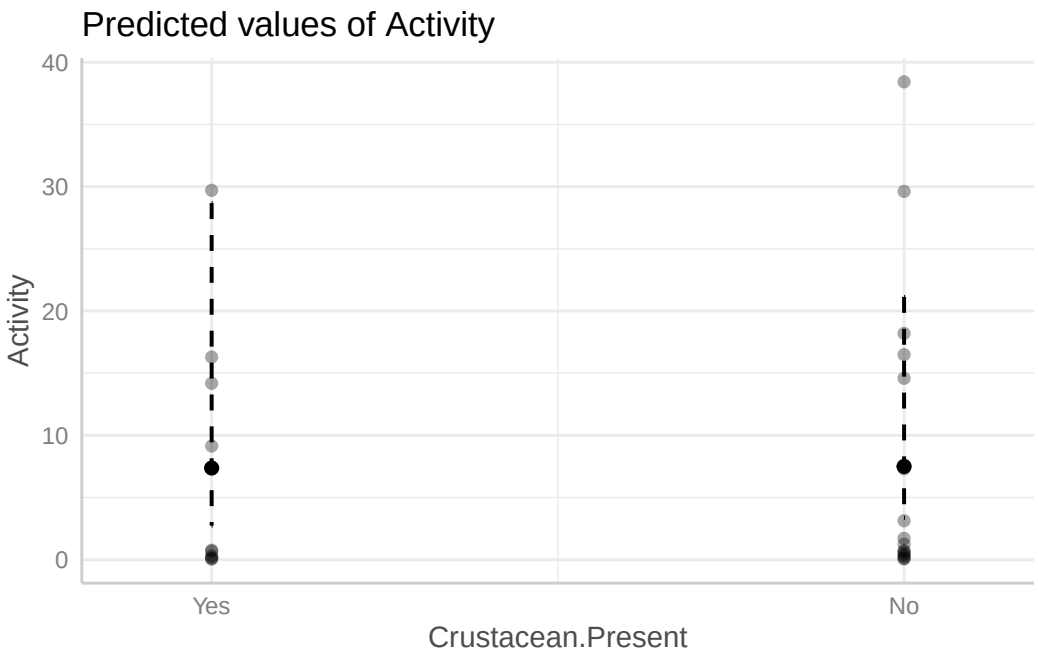



```r
plot(Lipase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.

$GI_tract_location
```



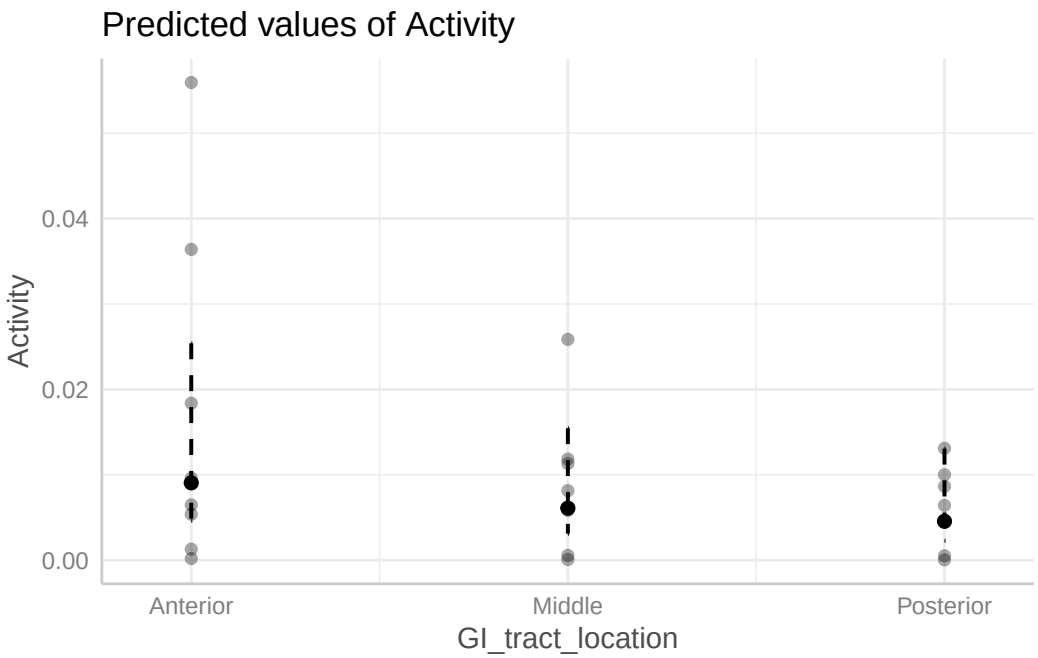

```
$Crustacean.Present
```



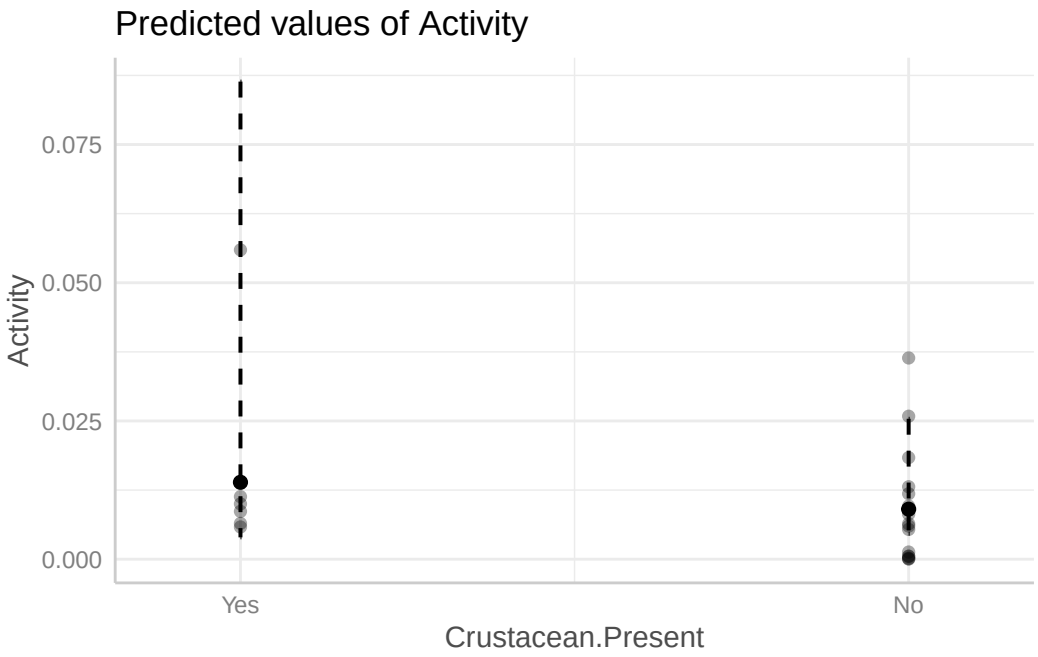
```r
plot(NAGase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```

```
$GI_tract_location
```

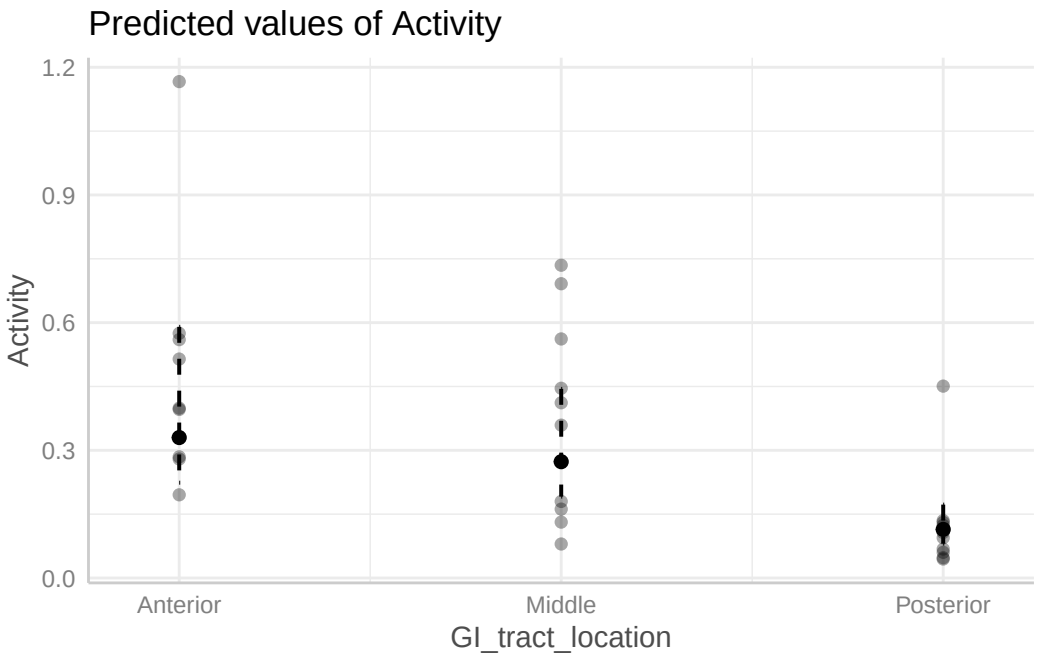



```
$Crustacean.Present
```

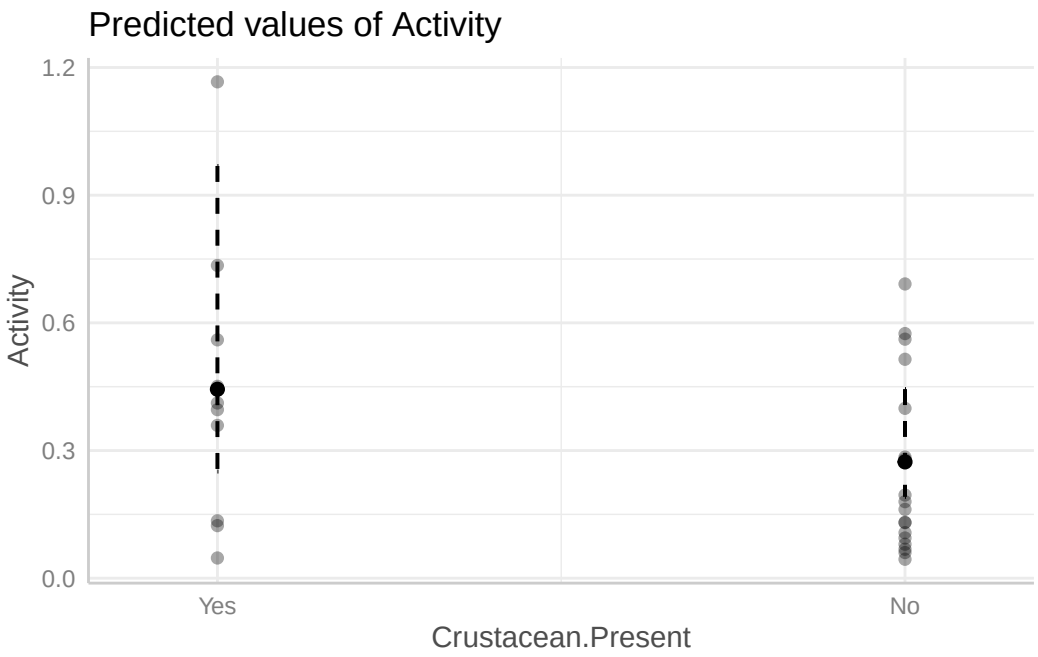



```r
plot(APase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.

$GI_tract_location
```



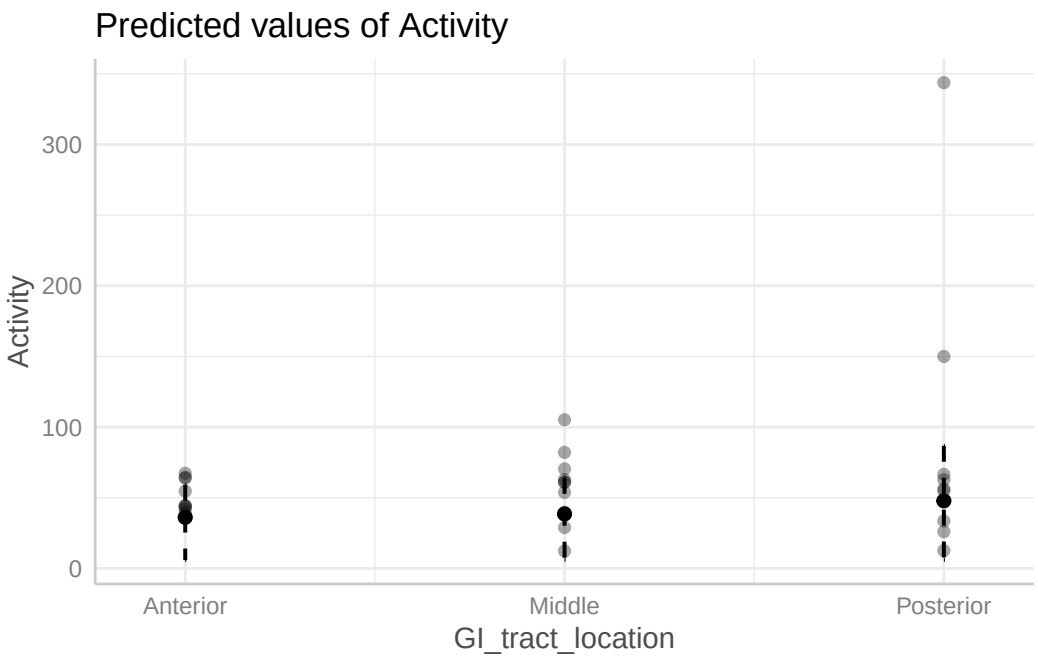

```
$Crustacean.Present
```



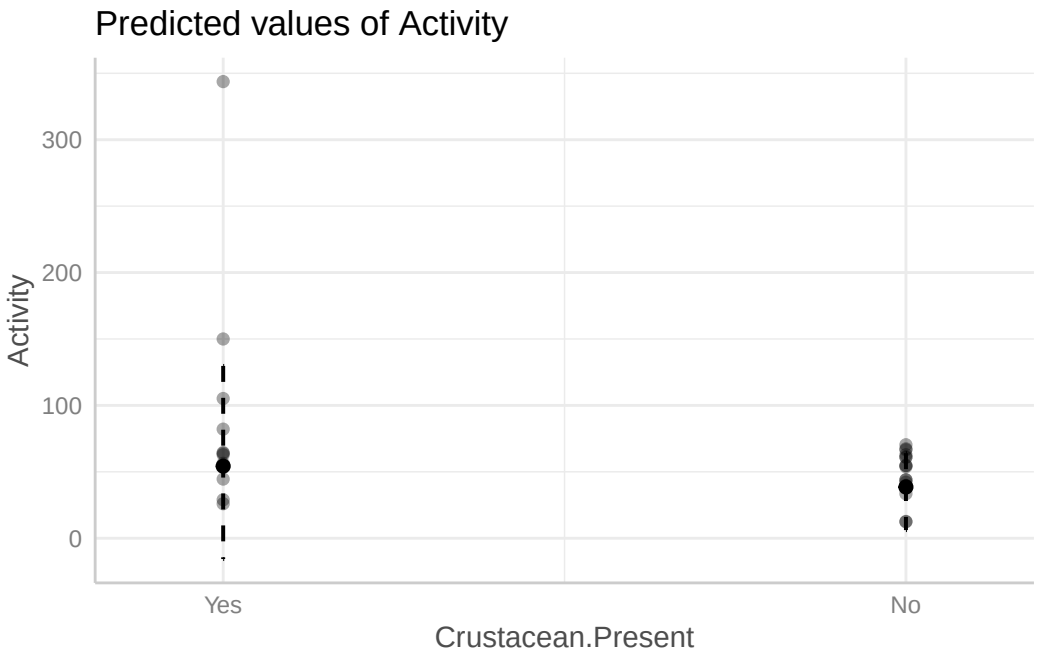
```r
plot(Trypsin_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```

```
$GI_tract_location
```

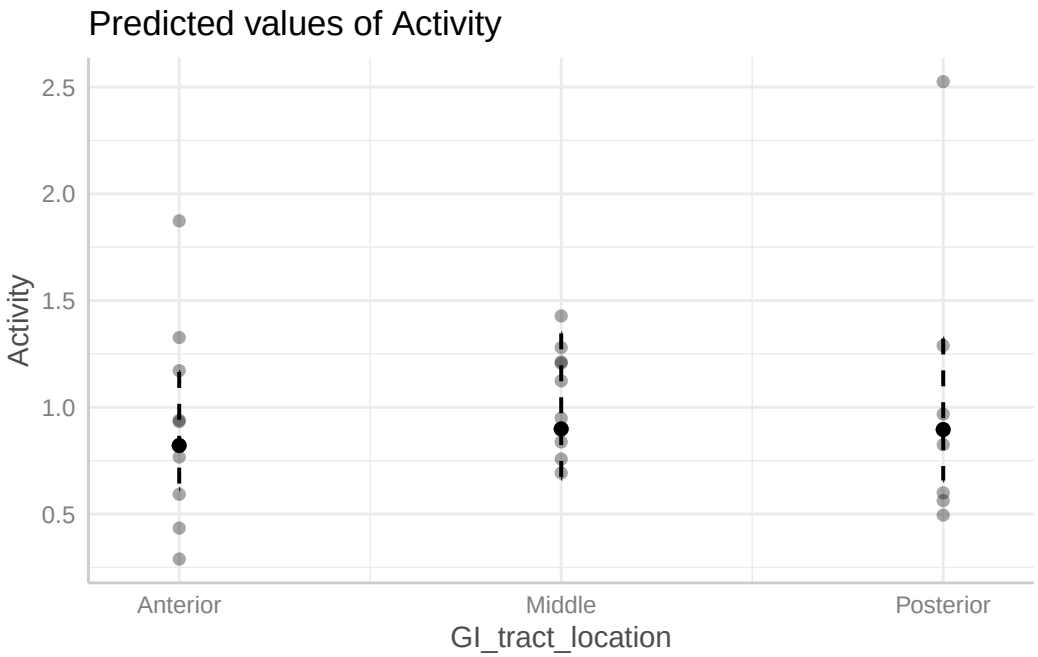



```
$Crustacean.Present
```

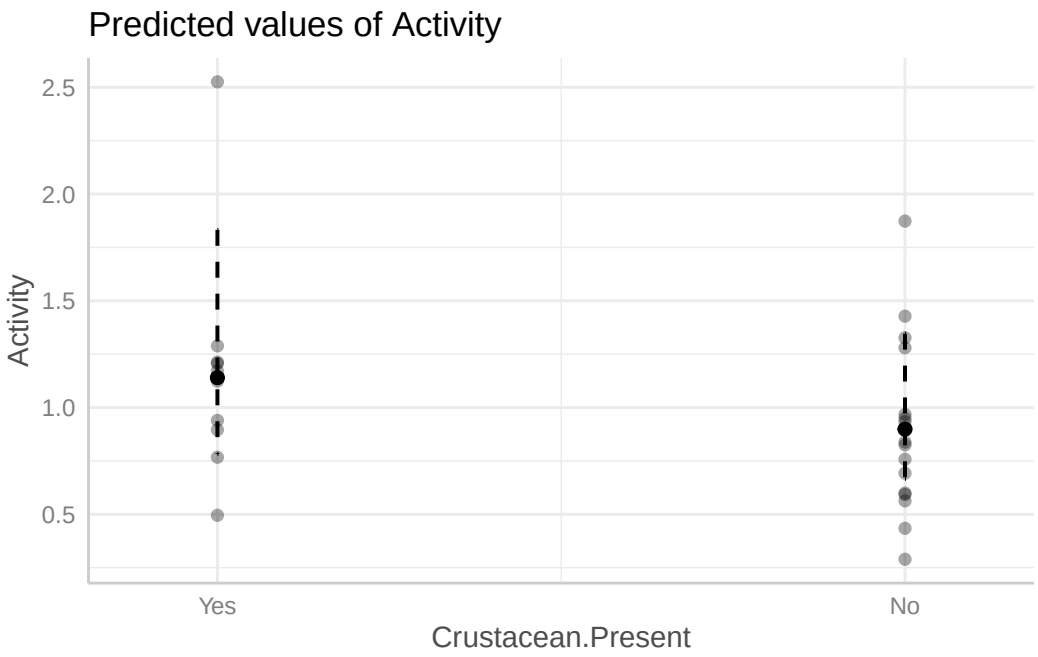



```r
plot(A.peptidase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.

$GI_tract_location
```




```
$Crustacean.Present
```



Building Models for stomach enzymes and showing results

Note: because the data set is really small, we will be creating priors for it. We will also be using the map2 function so that we only have to run the code for the model once and it loops through all the enzymes individually. We want to make sure the enzymes run separately because we are looking at them independently.

```
dir.create("output")
```

```
Warning in dir.create("output"): 'output' already exists
```

```
grouped <- stomach_long %>%
  group_by(Enzyme)

s.enzyme_data  <- group_split(grouped)
s.enzyme_names <- group_keys(grouped)$Enzyme

# Fit model for each enzyme
models.s <- map2(
  s.enzyme_data,
  s.enzyme_names,
  ~ {    # Calculate enzyme-specific prior center
    priors_s <- c(
      prior(normal(log(mean(.x$Activity)), 1), class = "b"),
      prior(exponential(1), class = "sd"),
      prior(gamma(2, 0.1), class = "shape")
    )
    brm(
      data = .x,
      family = Gamma,
      Activity ~ 0 + GI_tract_location + Crustacean.Present +
        (1|ID),
      control = list(adapt_delta = .9),
       iter = 2000, warmup = 1000, chains = 4, cores = 8,
      file = paste0("output/m.stomach_", .y)
    )}
)

names(models.s) <- s.enzyme_names
```

```r
print(models.s, digits = 3)
```

```
$Chitinase

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
careful when analysing the results! We recommend running more iterations and/or
setting stronger priors.

Warning: There were 2184 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 19)
  Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 3000

Multilevel Hyperparameters:
~ID (Number of levels: 9)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)  116.378    98.495    3.958  376.441 1.072       42       39

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAntrum  180.164    94.128   28.732  390.481 1.127       16
GI_tract_locationBody    214.312    96.869   46.852  408.463 1.118       17
GI_tract_locationFundus  181.530   100.183   24.067  436.415 1.146       14
Crustacean.PresentYes   -112.557   100.525 -365.128   59.286 1.081       28
                        Tail_ESS
GI_tract_locationAntrum       28
GI_tract_locationBody         23
GI_tract_locationFundus       23
Crustacean.PresentYes         13

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    0.653     0.249    0.313    1.267 1.011      141      248

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
```

```
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).

$Pepsin

Warning: There were 563 divergent transitions after warmup. Increasing
adapt_delta above 0.9 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: gamma
  Links: mu = inverse
Formula: Activity ~ 0 + GI_tract_location + Crustacean.Present + (1 | ID)
   Data: .x (Number of observations: 21)
  Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 3000

Multilevel Hyperparameters:
~ID (Number of levels: 10)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.152     0.138    0.005    0.514 1.003     1068     1178

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS
GI_tract_locationAntrum    1.057     0.345    0.487    1.804 1.003      810
GI_tract_locationBody      0.666     0.239    0.275    1.177 1.003      880
GI_tract_locationFundus    3.704     1.560    1.441    7.379 1.006     1089
Crustacean.PresentYes     -0.289     0.283   -0.887    0.245 1.003      952
                        Tail_ESS
GI_tract_locationAntrum      796
GI_tract_locationBody        981
GI_tract_locationFundus      774
Crustacean.PresentYes        901

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape    1.052     0.317    0.519    1.775 1.004     1408     1055

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
Chitinase_pred <- predict_response(models.s[[1]]) #, bias_correction = TRUE)
```

```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```r
Pepsin_pred <- predict_response(models.s[[2]]) #, bias_correction = TRUE)
```

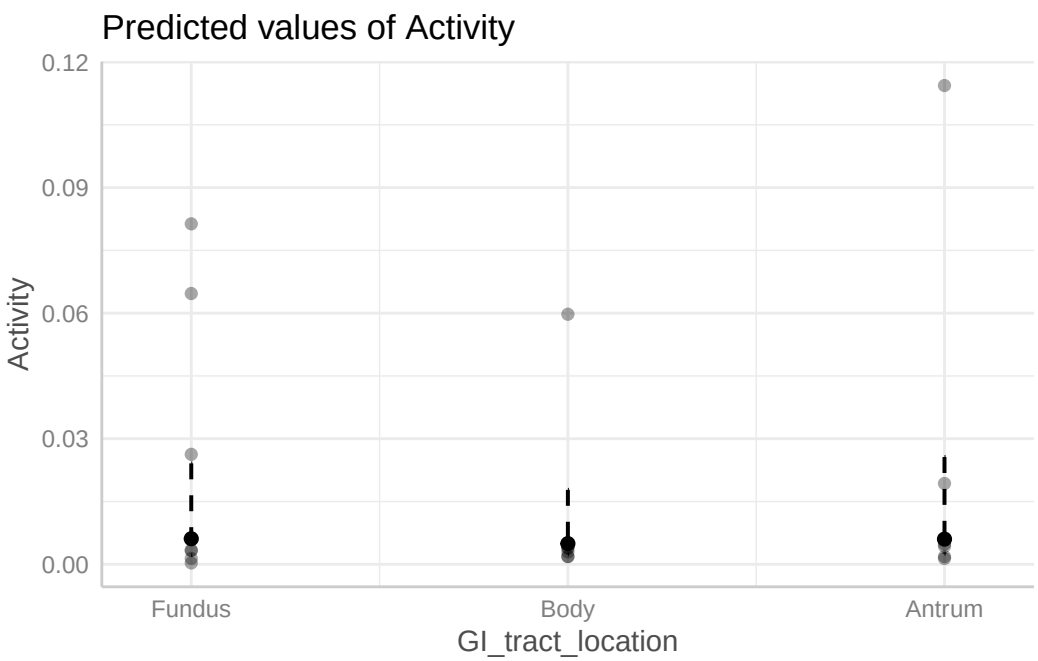
```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `GI_tract_location`
Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Crustacean.Present`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

Plotting the model predictions

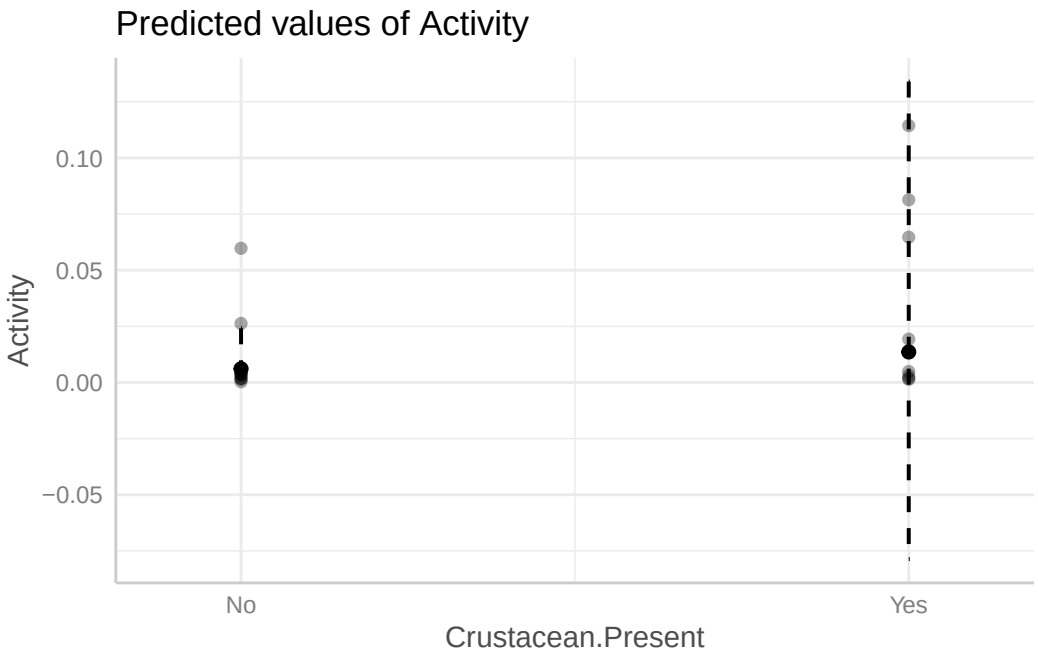
```
plot(Chitinase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```

```
$GI_tract_location
```

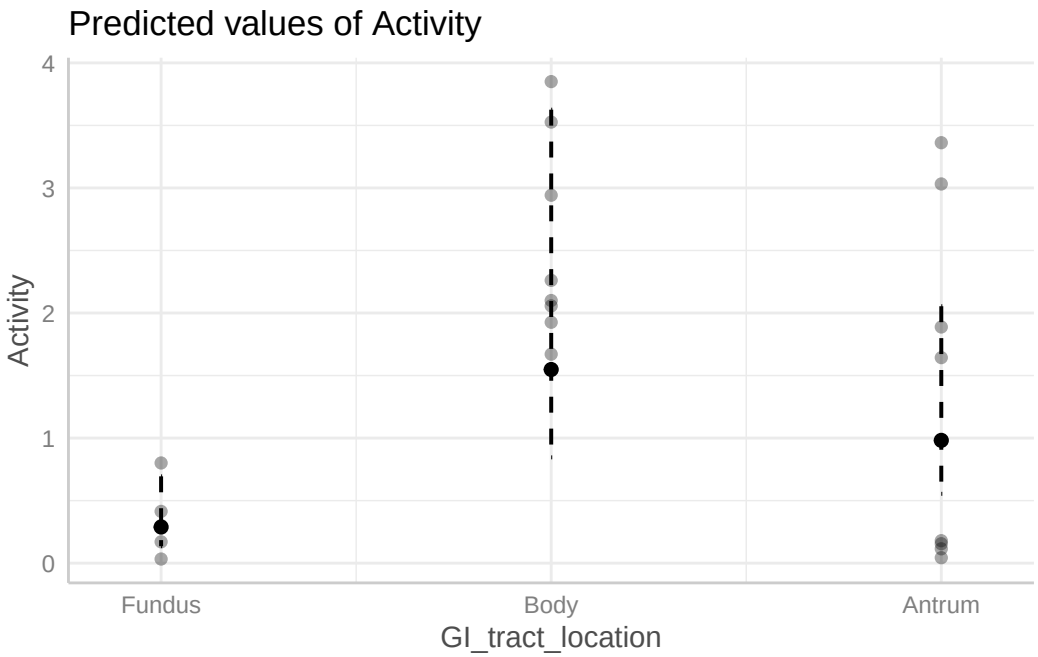



```
$Crustacean.Present
```

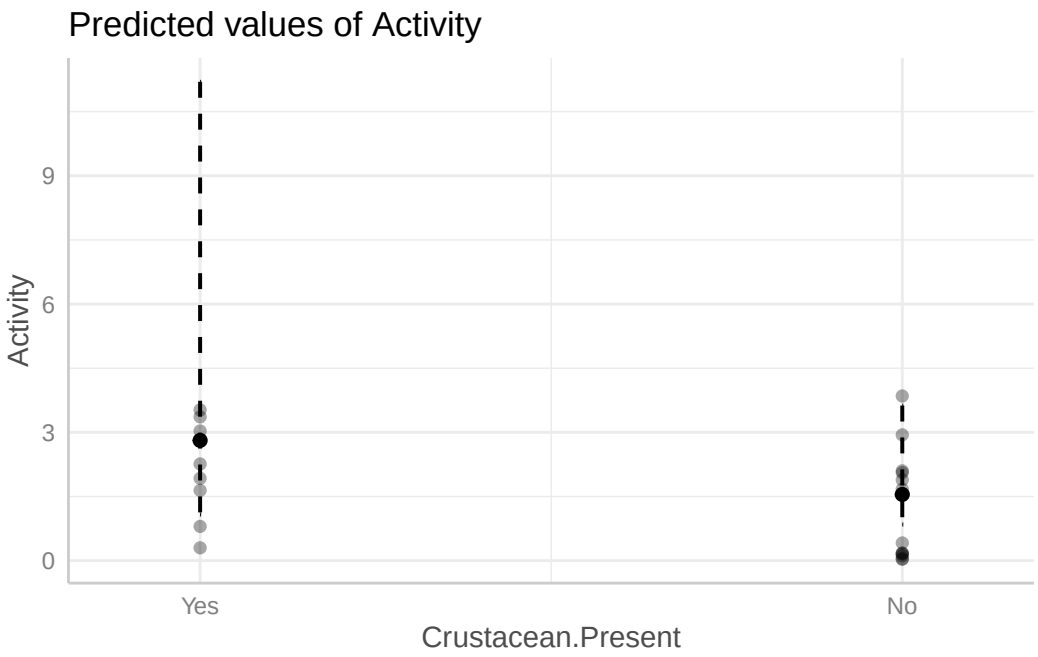



```r
plot(Pepsin_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.

$GI_tract_location
```




```
$Crustacean.Present
```

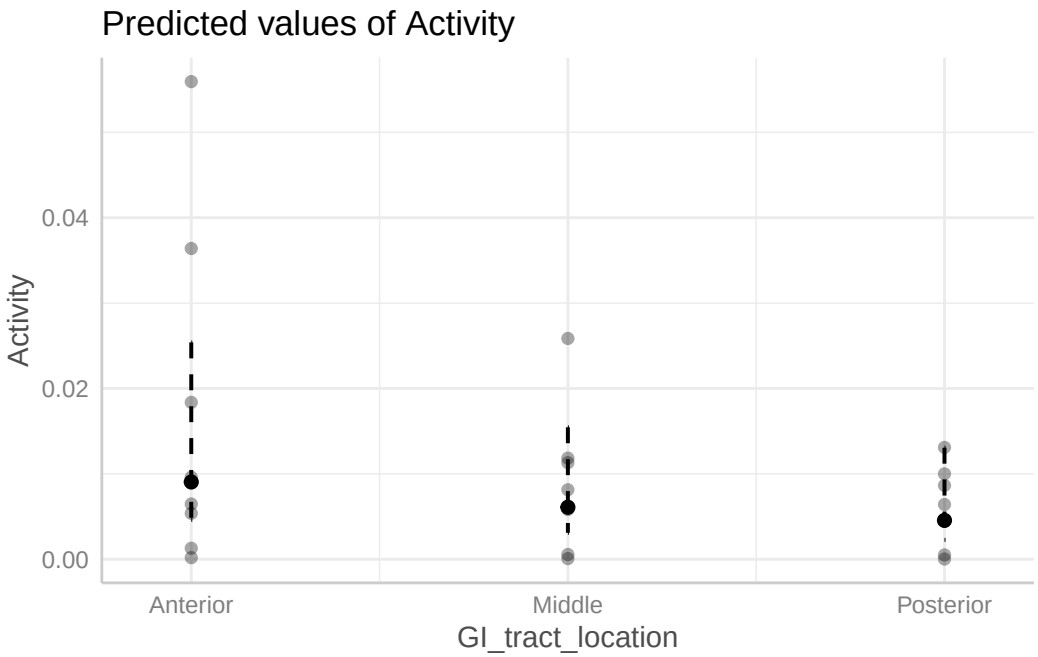


Plotting the results for the chitin degrading questions to see how the presence of crustacean prey interact with the activity and GI tract location

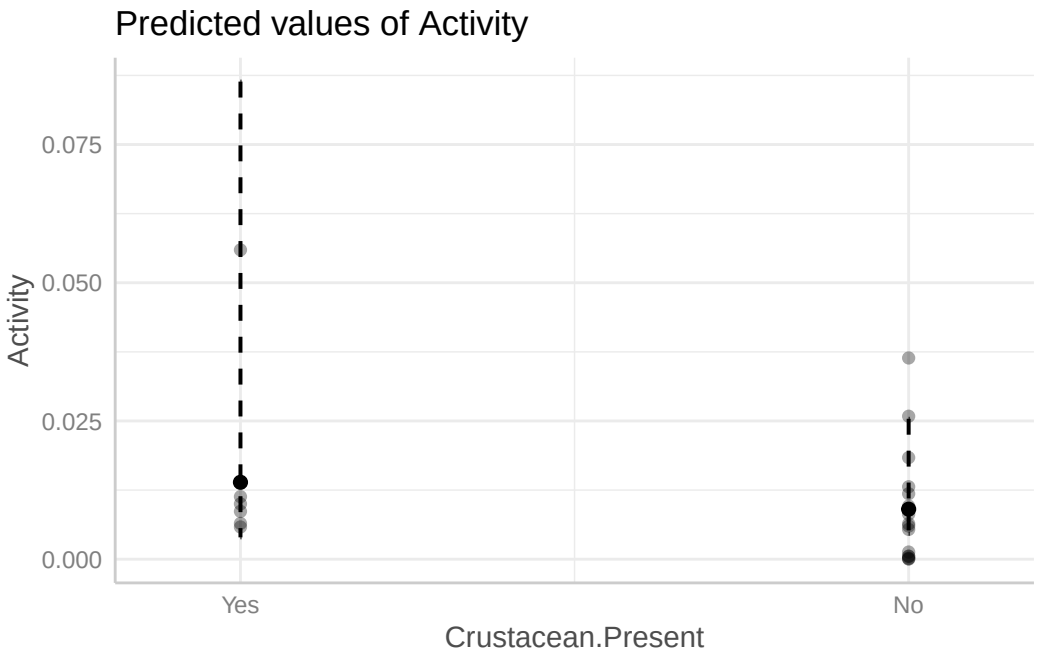
```
plot(NAGase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```

```
$GI_tract_location
```

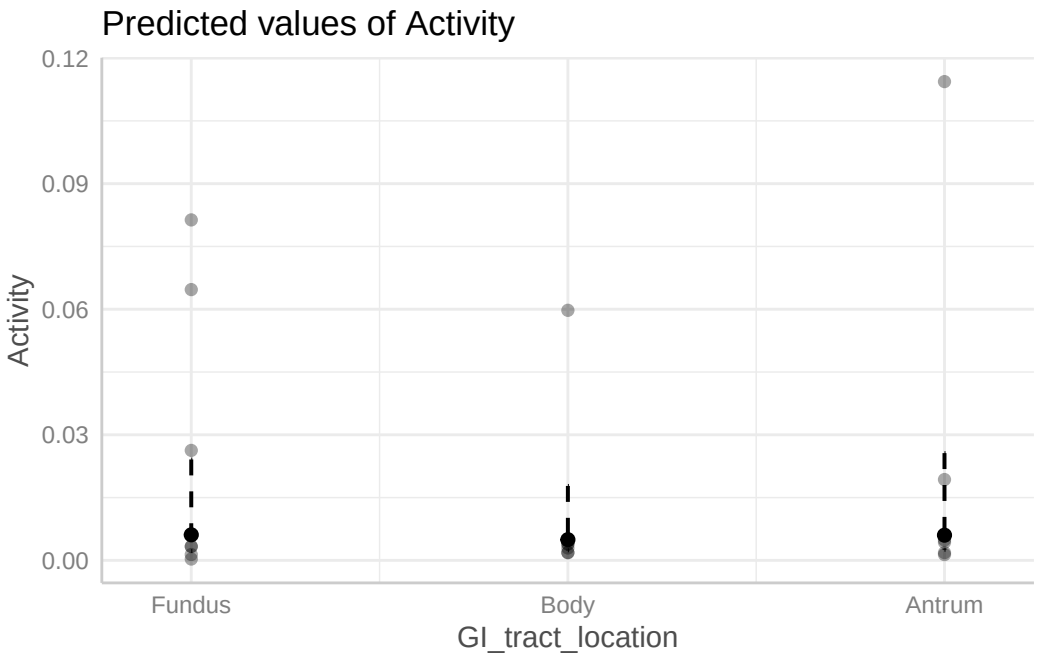



```
$Crustacean.Present
```

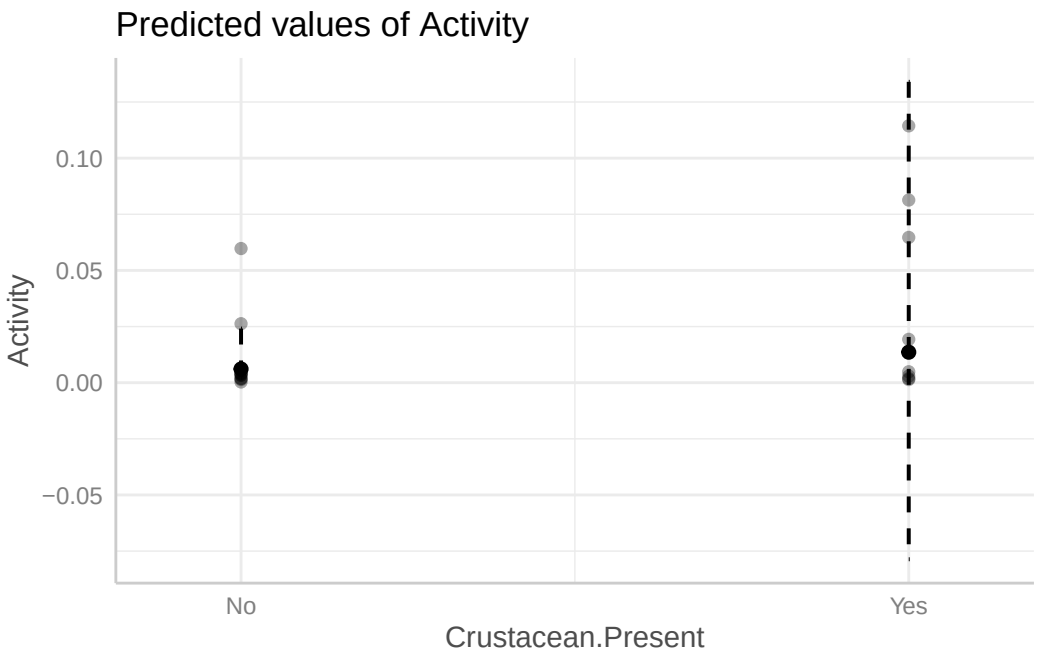



```r
plot(Chitinase_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.

$GI_tract_location
```



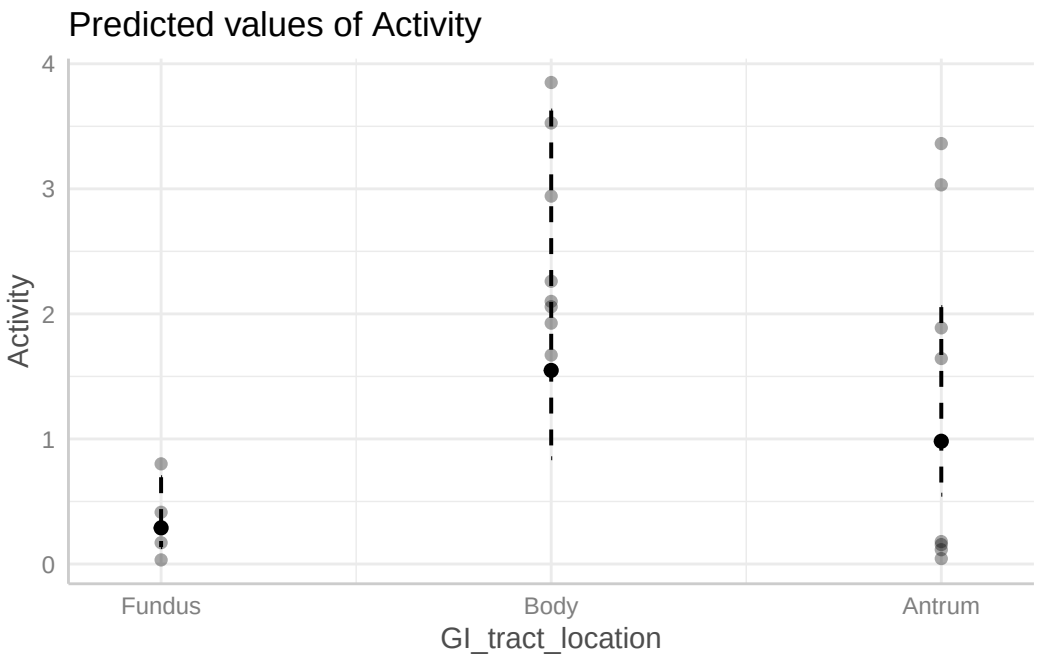

```
$Crustacean.Present
```



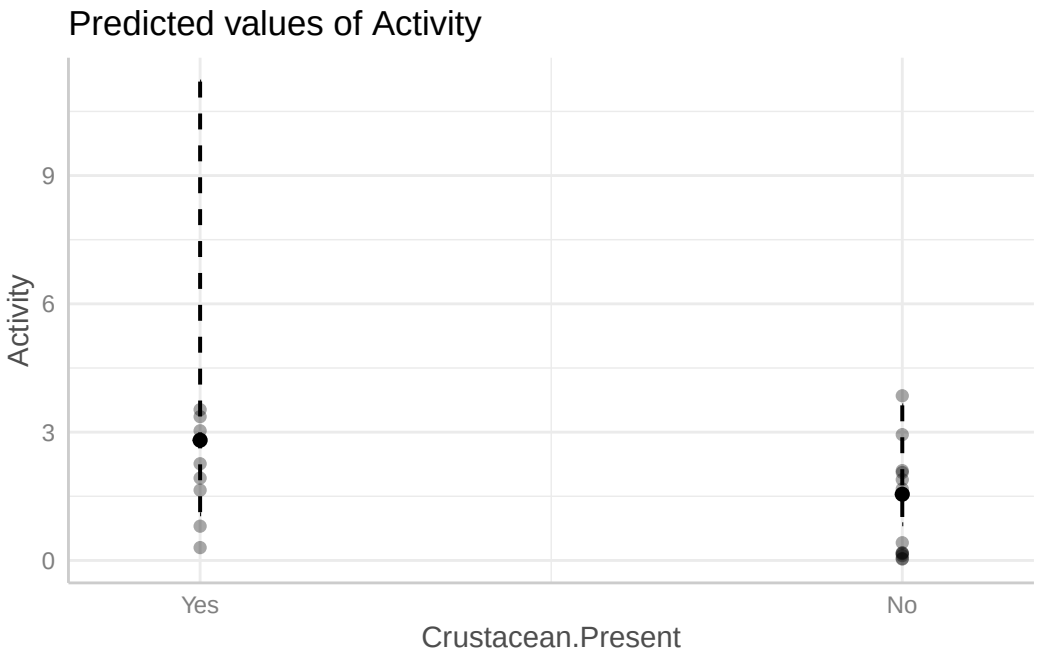
```r
plot(Pepsin_pred, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```

```
$GI_tract_location
```




```
$Crustacean.Present
```


Predicted values of Activity
Activity
9
6
3
0
Yes
No
Crustacean.Present